Mahesh Jivaji Shriram

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Professional Summary

AI Engineer with experience building production-grade **LLM-powered applications, Agentic AI systems, and retrieval-augmented solutions**. Experienced in designing **multi-agent workflows, AI-assisted engineering systems, MCP integrations and AI automation pipelines** for enterprise software engineering use cases. Hands-on exposure to **LLM fine-tuning, knowledge distillation, model quantization, embeddings, semantic search, and model evaluation**, with a strong foundation in **Machine Learning, NLP, PyTorch, TensorFlow and Python**. Proven experience taking AI solutions from experimentation to production with measurable engineering impact.

Technical Skills

- **Programming Language:** Python, C++, Javascript
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, NLP, Feature Engineering, Hyperparameter Tuning, Model Evaluation, MLflow
- **LLM Engineering:** LLM Fine-Tuning, LoRA, QLoRA, HuggingFace, Transformers, Knowledge Distillation, Model Quantization, Embeddings, Semantic Search, RAG, LLM Evaluation, Prompt Engineering
- **Agentic AI:** LangChain, LangGraph, Ollama, Multi-Agent Systems, MCP, Agentic Workflows, AI Hooks, Human-in-the-Loop (HITL), Guardrails
- **Backend & Infrastructure:** REST APIs, FastAPI, GitHub Actions, CI/CD Pipelines, PostgreSQL, pgvector, MySQL, Firebase, Vector Databases

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Experience

Harbinger Group — July 2025 – Present

AI Engineer

AI-Assisted Engineering Initiative

- Designed and orchestrated **AI-driven software engineering workflows** across code generation, auditing, validation, and repository integration to improve engineering consistency and delivery speed.
- Implemented spec-driven development workflows (SpecKit) to standardize and improve the reproducibility of AI-generated code.
- Enhanced GitHub-based CI/CD pipelines using agentic workflows for automated validation and code quality enforcement.
- Implemented **pre-tool and post-tool guardrails** to validate tool inputs, enforce coding standards using AI Hooks, and verify AI-generated outputs before repository integration.

Bug-Fixing Agent – AI-Powered Defect Resolution System

- Designed and built a **multi-agent system** for autonomous **software defect investigation, root-cause analysis, fix generation, and validation** across **.NET and Angular** applications.
- Automated **end-to-end bug resolution** using **Human-in-the-Loop (HITL)**, **browser-assisted validation**, and **self-healing agentic workflows** with automated knowledge-base updates.
- Integrated **Atlassian & GitHub MCP** for contextual issue and code retrieval, and **CodeGraph** for impact analysis and **token-efficient code understanding**.
- Implemented **pre-tool and post-tool guardrails** for **execution safety, fix validation**, and controlled automated remediation.

- Achieved **84% success across 42 defects**, saving **207+ engineering hours** (˜**4.9 hours per issue**).

DeepPulse - AI-Based Employee Feedback System

- Engineered an **LLM-powered feedback system** using **LangChain, RAG, and vector search** to generate personalized employee insights.
- Developed **context-aware questionnaire generation** using employee profiles, performance history, and lifecycle events.
- Integrated **PostgreSQL (pgvector)** for **embedding-based semantic retrieval** and context enrichment.
- Built **prompt orchestration pipelines** combining retrieved knowledge, employee context, and organizational objectives.
- Designed an **end-to-end AI pipeline** covering data ingestion, embedding generation, retrieval, prompt construction, and automated feedback generation.

Hartaraf Nov 2024 – Feb 2025

Software Engineer Intern (Remote)

- Developed an **intent classification and NLP pipeline** for a Flutter-based platform, enabling natural language service discovery for rural users with limited digital literacy.
- Implemented a **Speech-to-Text (STT) and Text-to-Speech (TTS)** pipeline to support voice-based queries, expanding accessibility for non-text users.
- Integrated **RAG-based service matching** to return contextually relevant local service providers based on user queries, powered by Firebase backend and REST APIs.

Independent Projects

ModelForge – LLM Fine-Tuning, Distillation & Quantization System

- Built a **model adaptation system** for **LLM fine-tuning, knowledge distillation, and model quantization** using compact instruction-tuned language models.
- Implemented **Supervised Fine-Tuning (SFT), LoRA/QLoRA, response distillation, and logit distillation** for model adaptation and compression.
- Evaluated base, fine-tuned, distilled, and quantized models across **model quality, inference latency, and memory footprint**.
- Implemented **model quantization** to reduce model size and memory footprint for resource-constrained inference environments.

ModelLens – Custom Benchmarking for Fine-Tuned Models

- Built an **automated LLM evaluation system** using **RAGAS and DeepEval** to assess faithfulness, answer relevance, context precision, and hallucination-related failure modes.
- Designed **domain-specific evaluation suites** and a **reproducible benchmarking harness** to compare base, fine-tuned, and quantized models across accuracy, generation quality, and inference latency.
- Integrated **Weights & Biases (W&B)** for experiment tracking, metric visualization, and **regression detection**, establishing an **evaluation-driven quality gate** for model promotion.

Education

Bachelor of Engineering in Artificial Intelligence and Data Science

PES Modern College of Engineering, Pune August 2021 – June 2025

Cumulative GPA: 8.2